Boolean Algebra and Logic Gates- Part 1

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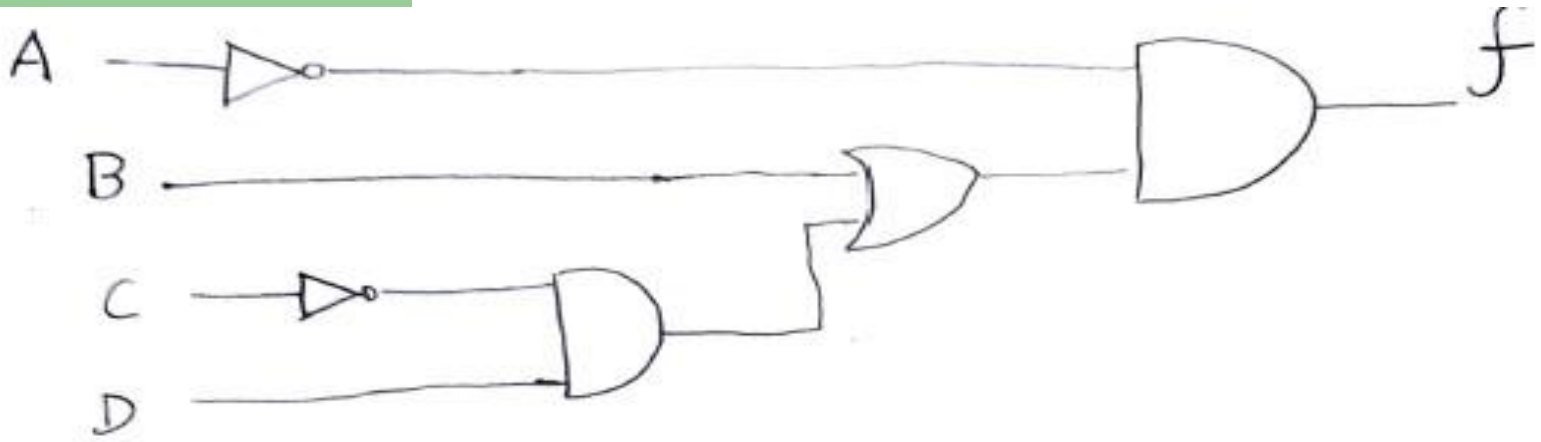
Review

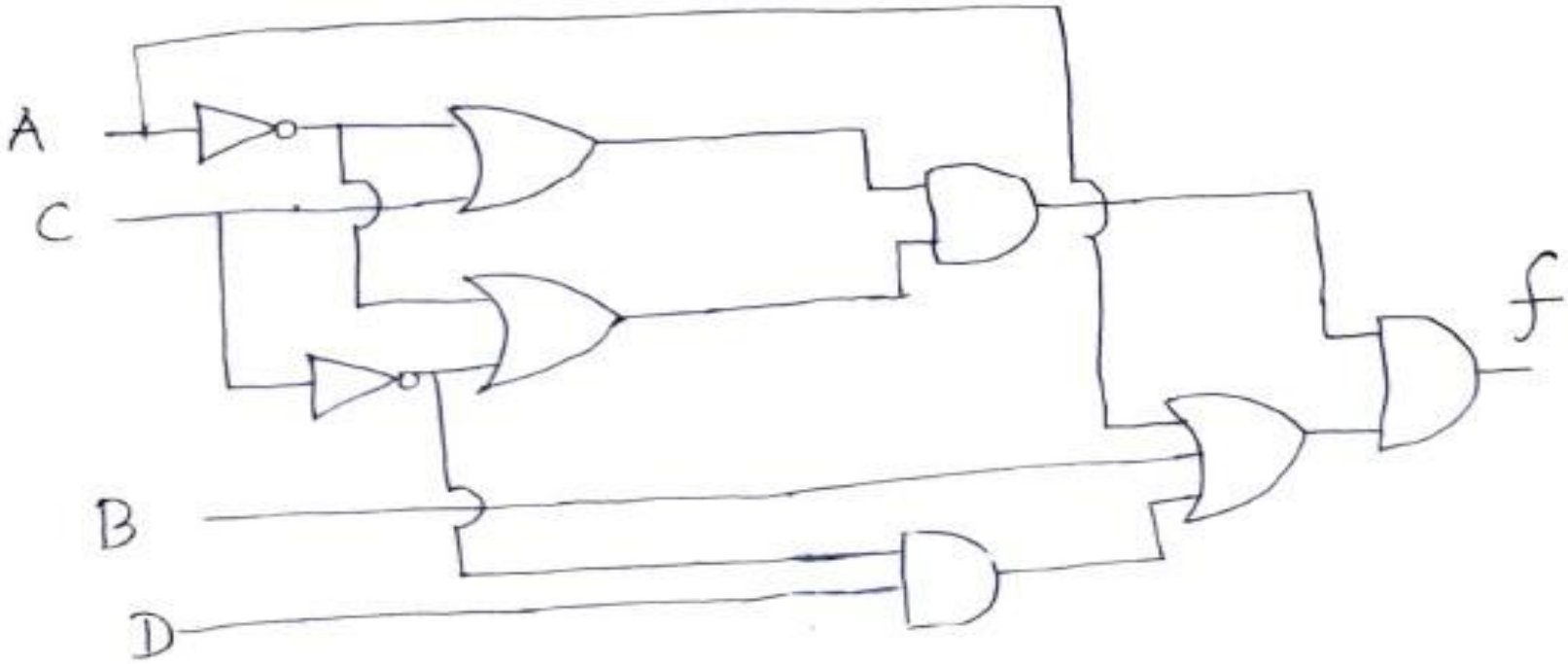
- We have so far discussed about number systems.
- We will now discuss something called Boolean (or switching) algebra.

Motivation

- When we want to design digital systems, a key consideration is the cost of the system.
- Alternatively, one may have other objectives such as high speed or low power consumption.
- We need to see if we can have a “simple” circuit to help us realize our goals.

Which circuit is better ?





Remarks on the circuits

- Both the circuits take the same set of inputs (A,B,C and D)
- It can be shown that both produce the same output f (can verify this by physically giving inputs A, B, C and D via a function generator and observing the output via an oscilloscope).
- We would prefer the circuit at the top (Why ?)

What does this have to do with Boolean algebra ?

- We can show that the outputs will be the same mathematically too !
- In particular, certain postulates (axioms) and theorems can be written and applied to show that the “expressions” for f in terms of A,B,C and D for the two circuits are equivalent.
- We will list the axioms and theorems and use them to show that f is the same.

Before proceeding with the axioms …

- Boolean algebra and binary logic are related.
- In particular, basic logic gates in circuits realize certain operations.
- The key operations are AND, OR and NOT.
- AND operation is denoted by **.** (dot symbol) while the OR operation is denoted by + (plus sign)
- The NOT operation is represented by ‘ (prime) or by a line (called overbar) just above a variable.

Very important !

- Note that dot and plus in binary logic are not the same as multiplication and addition in binary arithmetic in general.
- For example, 1 + 1 in binary logic is 1 while 1 + 1 in binary arithmetic is 10.
- In binary arithmetic, we often have many digits in a number while in binary logic, we have logic variables which take only values of 0 or 1.
- With this background, we will proceed to axioms of Boolean algebra.

Axioms of Boolean algebra

- Boolean algebra is an algebraic structure defined by a set of elements B together with two binary operators, denoted by + and . such that the following are satisfied:
- Element 0 is an identity element with respect to + while 1 is an identity element with respect to . That is,

  (a) x + 0 = x (b) x .1 = x
- The structure is commutative with respect to + and . (That is, (a) x + y = y + x (b) x . y = y . x). x.y may be written as xy
- Operator . is distributive over +
- For every element x in B, there exists an element x' in B such that x + x' = 1 and x. x' = 0

Main theorems of Boolean algebra

- We state the theorems first.
- Theorem 1: (a) x + x = x (b) x . x = x
- Theorem 2: (a) x + 1 = 1 (b) x . 0 = 0
- Theorem 3: (x')' = x
- Theorem 4: (a) x + xy = x (b) x(x + y) = x
- Theorem 5 (DeMorgan); we omit the dot

  (a) (x + y)' = x'y' (b) (xy)' = x' + y'

Outline of the proofs

- All the theorems can be proved using the axioms (postulates). For example, $x + x =$

  $(x + x) \cdot 1 = (x + x)(x + x') = x + 0 = x$

  Also, $x \cdot x = xx + 0 = xx + xx' = x(x + x') = x$

  Further, $x + 1 = 1 \cdot (x + 1) = (x + x')(x + 1)$

  $= x + x' \cdot 1 = x + x' = 1$

Duality

- Duality principle (or just duality) is an important property of Boolean algebra.
- To obtain the dual of an algebraic expression, simply interchange OR and AND operators and replace 1's by 0's and 0's by 1's.
- This can be used to show that x.0 = 0 from

  x + 1 = 1

Are there other ways to prove these ?

Yes, there is something called truth table which we will study a little later. This can be used to show equivalence of different Boolean expressions.

The DeMorgan's Laws can be established easily using truth tables.

Let’s get back to the logic circuits shown earlier (please see slide 4)

- Given the axioms and theorems of Boolean algebra, we are in a position to apply them to show the equivalence of the two circuits.
- The detailed proof is shown next.

Equivalence of circuits in slide 4

The top circuit yields $f = \bar{A}(B + \bar{C}D)$

The bottom circuit yields $f = (\bar{A}+C)(\bar{A}+\bar{C})(A+B+\bar{C}D)$

We can show that $(\bar{A}+C)(\bar{A}+\bar{C})(A+B+\bar{C}D)$ is same as $\bar{A}(B+\bar{C}D)$ as follows.

$$(\bar{A}+C)(\bar{A}+\bar{C})(A+B+\bar{C}D) = (\bar{A}\cdot\bar{A} + \bar{A}\cdot\bar{C} + C\bar{A} + C\cdot\bar{C})(A+B+\bar{C}D)$$

$$= (\bar{A} + \bar{A}\cdot\bar{C} + C\bar{A} + 0)(A+B+\bar{C}D)$$

$$= \bar{A}(1 + \bar{C} + C)(A+B+\bar{C}D) = \bar{A}(A+B+\bar{C}D)$$

$$= \bar{A}\cdot A + \bar{A}\cdot B + \bar{A}\bar{C}D$$

$$= \bar{A}(B+\bar{C}D)$$

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- J. Wakerly, Digital Design – Principles & Practices, Pearson Education, 2008